

The diagram illustrates the control system for a Permanent Magnet Synchronous Motor (PMSM), divided into Hardware and Software blocks.

Hardware Blocks:

- 1. Permanent Magnet Synchronous Motor:** The motor itself, receiving three-phase currents (IA, IB, IC) and providing feedback signals (Iα, Iβ, Vα, Vβ, Id, Iq, ω_{mech}).
- 2. 3-Phase Bridge:** The power electronic converter that receives reference voltages (Vα, Vβ) and provides the three-phase currents to the motor.

Software Blocks:

- 3. A, B:** A coordinate transformation block that converts three-phase currents (Iα, Iβ) into two-phase currents (Id, Iq).
- 4. d, q:** A coordinate transformation block that converts two-phase currents (Id, Iq) into three-phase currents (Iα, Iβ).
- 5. Estimator:** A block that estimates the motor's speed and angle based on feedback signals (Iα, Iβ, Vα, Vβ, Id, Iq, ω_{mech}).
- 6. PI:** Proportional-Integral controllers for the reference speed (ω_{ref}) and reference d-axis current (Id_{ref}).
- 7. Flux Weakening:** A block that generates a reference d-axis current (Id_{ref}) based on the mechanical speed (ω_{mech}).
- 8. SVM:** Space Vector Modulation block that generates reference voltages (Vα, Vβ) based on the reference d-axis current (Id_{ref}) and the estimated angle.

Control Loop:

- The reference speed (ω_{ref}) is compared with the estimated speed to generate an error signal.
- This error signal is processed by a PI controller (6) to produce a reference d-axis current (Id_{ref}).
- The reference d-axis current (Id_{ref}) is compared with the estimated d-axis current (Id) to generate an error signal.
- This error signal is processed by a PI controller (6) to produce a reference q-axis current (Iq_{ref}).
- The reference q-axis current (Iq_{ref}) is compared with the estimated q-axis current (Iq) to generate an error signal.
- This error signal is processed by a PI controller (6) to produce a reference voltage (Vα, Vβ).
- The reference voltages (Vα, Vβ) are processed by the SVM block (8) to generate the reference three-phase currents (IA, IB, IC).
- The reference three-phase currents (IA, IB, IC) are provided to the 3-Phase Bridge (2) to drive the PMSM (1).

- ## Hardware Blocks

1. Permanent Magnet Synchronous Motor.
2. 3-Phase Bridge – rectifier, inverter and acquisition and protection circuitry.
3. software (run by the device).
4. Clarke forward transform.
5. Park forward and inverse transform.
6. Angle and speed estimator.
7. Proportional integral controller.
8. Flux weakening.
9. Space vector modulation.